TEXORA · LECTURE SHEET

Human Physiology

The body the clothing serves: thermoregulation and the heat-balance equation, metabolic rate and body surface area, sweating and its cooling arithmetic, blood-flow control, and the strain limits that protective and comfort design must respect.

AM-4103 · CLOTHING COMFORT & SPECIAL APPAREL PRODUCTS · TOPIC 5 · LEVEL 4 TERM 1

1. The regulated body

Humans are homeotherms: the **hypothalamus** holds core temperature near **37 °C** (normal range about 36.5–37.5 °C) by balancing heat production against heat loss, using sensors in the core and the skin (the receptors of Topic 1 are its instruments). The controlled variable is core temperature; the manipulated variables are **skin blood flow** (vasodilation pours heat to the surface, vasoconstriction insulates the core), **sweating** (evaporative cooling), **shivering** (emergency heat production) and behaviour (seeking shade, removing clothing — always the strongest response, and the one bad garment design provokes against PPE). Skin temperature, by contrast, is allowed to wander: comfort sits at about **33–35 °C** mean skin temperature, and it is skin temperature the wearer actually feels.

2. The heat-balance equation

$$M - W = R + C + K + E + S \quad [\text{W or W/m}^2]$$

TERM	MEANING	CLOTHING'S INFLUENCE
M	Metabolic heat production (activity-driven).	None — set by the task; the design input.
W	External mechanical work (small for most jobs; most energy becomes heat).	Negligible.
R, C, K	Dry heat exchange: radiation, convection, conduction.	Governed by insulation (R_{ct}, tog/clo), air permeability, fit, wind.
E	Evaporative loss (sweat + respiration).	Governed by R_{et}/MVTR — the protective-clothing bottleneck.
S	Heat storage: S > 0 means core temperature rising.	The failure term: sustained S > 0 is heat strain.

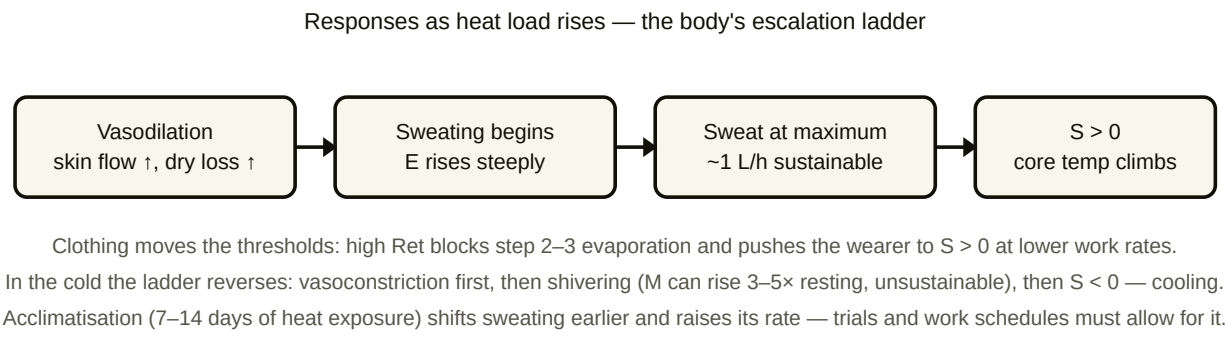


Figure 1 — The thermoregulatory escalation ladder. Comfort design keeps the wearer on the left-hand steps; protective design must budget time on the right-hand ones.

3. Metabolic rate and body surface area

Heat production is set by activity and is tabulated per unit skin area. The unit is the **met**: 1 met = **58.15 W/m²**, the resting seated rate. Body surface area is estimated by the Du Bois formula:

$$A = 0.007184 \times W^{0.425} \times H^{0.725} \quad (W \text{ in kg, } H \text{ in cm, } A \text{ in m}^2)$$

ACTIVITY	MET	W/M²	TOTAL FOR A = 1.85 M²
Sleeping	0.7	≈ 41	≈ 76 W
Seated at rest (1 met)	1.0	58	≈ 107 W
Light bench work / office	1.6	≈ 93	≈ 172 W
Walking 4 km/h, light industry	2.5	≈ 145	≈ 269 W
Heavy manual work	4.0	≈ 233	≈ 431 W

WORKED EXAMPLES — VERIFY EVERY NUMBER

(a) Surface area. W = 70 kg, H = 175 cm: $A = 0.007184 \times 70^{0.425} \times 175^{0.725}$ = **1.85 m²**.

(b) Heat to shed. The same person at 2 met produces 2 × 58.15 × 1.85 = **215 W**. At 20 °C in 1 clo, dry loss (Topic 1) is ≈ 151 W, so evaporation must remove ≈ 215 − 151 = **64 W** — a modest sweat rate, easily met by a breathable garment.

(c) Sweat arithmetic. Evaporating 1 g of sweat absorbs ≈ 2,430 J. Sweating 0.5 L/h evaporated fully gives 500 × 2,430 ÷ 3,600 = **338 W** of cooling — the body's largest heat-loss channel, and exactly the channel a high-Ret garment closes. Sweat that drips off unevaporated gives almost no cooling: efficiency of sweating, not its rate, is what clothing controls.

4. Strain: when balance fails

Heat strain is the physiological cost when S > 0 persists: core temperature rises, heart rate climbs to sustain skin blood flow (cardiovascular drift), and dehydration compounds both — a 2% body-mass water loss already degrades performance and judgement. Occupational guidance treats a sustained core temperature of about **38 °C** as the ceiling for acclimatised workers, with work–rest cycles and hydration prescribed beyond it; heat exhaustion and heat stroke (core > 40 °C, a medical emergency) mark the end of the scale. **Cold strain** mirrors it: extremity cooling kills dexterity (manual tasks fail before core temperature does), and frostbite threatens exposed skin in wind chill. Protective ensembles are therefore issued with **physiological budgets** — permitted exposure durations computed from M, climate and the garment's measured R_{ct}/R_{et} (ISO 7933 for heat, ISO 11079/IREQ for cold) — the direct application of this topic to the products of Topic 3.

5. Individual variation

Every number in this sheet is a population mean. Body size and composition change area-to-mass ratio; fitness raises sustainable sweat rate; age and medication blunt responses; women and men differ on average in sweat onset and distribution; acclimatisation state shifts every threshold. Standards handle this by testing with panels and designing to percentiles; engineers handle it by leaving margin — the wearer in the garment is never the average in the table.

Glossary

TERM	MEANING
Met	Unit of metabolic rate: 1 met = 58.15 W/m², the seated resting rate.
Vasodilation / vasoconstriction	Widening / narrowing of skin blood vessels to raise / cut dry heat loss.
Heat strain	The physiological response load (core temperature, heart rate, dehydration) when heat balance cannot be held.
Acclimatisation	Adaptation over 7–14 days of heat exposure: earlier, heavier sweating and lower strain at the same load.
IREQ / ISO 7933	Standards computing required clothing insulation (cold) and predicted heat strain (hot) from activity, climate and garment data.

Key points and summary

- Core temperature is regulated at ~37 °C; comfort lives at skin 33–35 °C; clothing serves the balance $M - W = R + C + K + E + S$.
- Metabolic rate (met) × Du Bois area gives the heat load: 1 met ≈ 107 W for a 1.85 m² adult.
- Evaporation is the biggest cooling channel (0.5 L/h ≈ 338 W) — and the one Ret restricts.
- Sustained core ≈ 38 °C is the occupational ceiling; protective work runs on physiological time budgets.

Practice

Answer in your own words; brief model answers follow.

1. Calculate the Du Bois area of a person of 60 kg and 165 cm, and their resting (1 met) heat production. (Use $A = 0.007184 \times W^{0.425} \times H^{0.725}$; you should get about 1.66 m².)
2. A worker at 3 met (A = 1.8 m²) wears clothing whose dry loss is 120 W and whose Ret limits evaporation to 150 W. Is heat balance held? If not, what is S and what follows?
3. How much cooling does evaporating 0.8 L of sweat in one hour provide?
4. Why does sweat dripping off the body represent failed thermoregulation even though sweat rate is high?
5. Why do heat-exposure limits shorten for unacclimatised workers?

Answer key

1. $A = 0.007184 \times 60^{0.425} \times 165^{0.725} \approx 0.007184 \times 5.69 \times 40.6 \approx 1.66$ m². Resting heat = 58.15 × 1.66 ≈ 97 W.
2. Production = 3 × 58.15 × 1.8 = 314 W. Losses = 120 + 150 = 270 W. S = 314 − 270 = +44 W — balance fails; storage is positive, so core temperature rises steadily and exposure must be time-limited (work–rest schedule) or the garment changed.
3. 800 × 2,430 ÷ 3,600 = 540 W.
4. Cooling comes only from evaporation at the skin. Dripped sweat carries away negligible heat but full water and salt loss — the wearer dehydrates without the benefit, so strain (heart rate, core temperature) still climbs.
5. Acclimatisation raises sweat capacity and brings its onset earlier, with lower cardiovascular cost. Without it the same load produces higher core temperature and heart rate sooner, so permitted durations are set shorter until 7–14 days of adaptation are complete.